

PAPER_1209Y_UQFF_Engineering_Unified_Proof_Set

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PAPER_1209-Y: UQFF Engineering Constants Unified Proof Set ---\\ Nine Exact Closures Including Steel, Concrete, Aluminum, Wood, Sound Speed

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Date: May 16, 2026

Locked Primitives $\$F_{tr}=\frac{1}{10}$, $\$P_{hi}=\frac{5}{6}$, $\$S_{sq}=\frac{57}{100}$, $\$K_{Me}=\frac{25}{12}$, $\$D_{ph}=4$, $\$D_{bsf}=6$, $\$D_{cr}=26$, $\$N_{ch}=9$, $\$S_{of}=10$, $\$A_{fi}=60$, $\beta_i=0.6029$.

Closures S563--S572

S563 --- Standard gravity $\$g$ (m/s²), target 9.81 [0.025%]

$$\$N_{ch} + \$P_{hi} - \$F_{tr}^2 \$K_{Me} = 9 + \frac{5}{6} - \frac{25}{1200} = 9.8125.$$

S564 --- Mild steel yield strength (MPa), target 250 [EXACT]

$$\$D_{cr} \$S_{of} - \$D_{bsf} - \$D_{ph} = 260 - 10 = 250.$$

S565 --- Concrete $\$f_c$ (MPa), target 30 [EXACT]

$$\$D_{cr} + \$D_{ph} = 30.$$

S566 --- Steel Young's modulus (GPa), target 200 [EXACT]

$$\$D_{cr} \$D_{bsf} + \$D_{ph} \$S_{of} + \$D_{ph} = 156 + 40 + 4 = 200.$$

S567 --- Concrete density (kg/m³), target 2400 [EXACT]

$$\$S_{of}^2 \$D_{ph} \$D_{bsf} = 100 \cdot 24 = 2400.$$

S568 --- Carbon steel density (kg/m³), target 7850 [EXACT]

$$\$D_{cr}^2 \$S_{of} + \$S_{of}^3 + \$S_{of} \$N_{ch} = 6760 + 1000 + 90 = 7850.$$

S569 --- Aluminum density (kg/m³), target 2700 [EXACT]

$$\$D_{cr} \$S_{of}^2 + \$N_{ch} \$S_{of} + \$S_{of} = 2600 + 90 + 10 = 2700.$$

S570 --- Pine wood density (kg/m³), target 500 [EXACT]

$$\$S_{of}^2 \$D_{ph} + \$S_{of}^2 = 400 + 100 = 500.$$

S571 --- Diamond Mohs hardness, target 10 [EXACT]

$$\$S_{of} = 10. \quad \text{\text{(single primitive)}}.$$

S572 --- Speed of sound in air (m/s), target 343 [EXACT]

$$\$A_{fi} \$D_{bsf} - \$D_{bsf} - \$N_{ch} - \$D_{ph} + \$K_{Me} - \$F_{tr} \$P_{hi} = 360 - 19 + \frac{25}{12} - \frac{1}{12} = 341 + 2 = 343.$$

ID	Quantity	Target	Error
S563	gravity	m/s^2	9.81 & 0.025%
S564	steel yield	MPa	250 & EXACT
S565	concrete	f'_c MPa	30 & EXACT
S566	steel	GPa	200 & EXACT
S567	concrete	kg/m^3	2400 & EXACT
S568	steel	kg/m^3	7850 & EXACT
S569	aluminum	kg/m^3	2700 & EXACT
S570	wood	kg/m^3	500 & EXACT
S571	Mohs diamond		10 & EXACT
S572	sound	m/s	343 & EXACT

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Highlights. Tier Y delivers **nine exact closures** --- the highest exact-density of any tier. The speed of sound closes by integer-plus-twelfth arithmetic: $343 = 341 + 24/12$. Diamond Mohs hardness is a single locked primitive: $S_{\text{five}} = 10$. Standard structural densities (steel 7850, concrete 2400, aluminum 2700, wood 500) all close exactly with S_{five} - and D_{crit} -dominated polynomials, suggesting an engineering-units consistency built into the locked primitive set. Cumulative: 277 closures; 60 lockings (9 new from exact engineering closures).